

API Reference

This document is the API reference for the agent marketplace backend.

The implementation is HTTP-first, versioned under `/v1` for domain routes, and uses FastAPI-style JSON bodies and status codes.

Common Mechanics

Base URLs

- Health routes: `/health`, `/health/live`, `/health/ready`
- Domain routes: `/v1/...`

Common Headers

- **Authorization: Bearer <jwt-or-api-key>** for protected routes that accept generic bearer auth
- **Idempotency-Key** for invoke requests and provider payout requests
- **X-Request-ID** for request correlation
- **PAYMENT-REQUIRED** on `402 Payment Required` responses
- **PAYMENT-SIGNATURE** on paid retry requests produced by the x402 client
- **PAYMENT-RESPONSE** on successful paid invoke responses
- The service uses **X-Request-ID** for request correlation and echoes it when present

Common Error Shape

Most route handlers raise FastAPI HTTP errors with JSON bodies shaped like:

```
{
  "detail": "human-readable message"
}
```

Authentication Matrix

- Public: health routes, discovery routes, quote creation, and auth nonce / verify / refresh
- Generic bearer auth: provider authoring, invoke, invocation reads, provider earnings, ledger, public payout reads, and admin routes
- JWT-only: API-key lifecycle, account self-service, wallet rotation, and provider payout execution

Health

GET `/health`

- Auth: none
- Request: no body, no parameters

- Success: 200 with {"status":"ok"}
- Errors: none expected in the normal path

GET /health/live

- Auth: none
- Request: no body, no parameters
- Success: 200 with {"status":"ok"}
- Errors: none expected in the normal path

GET /health/ready

- Auth: none
- Request: no body, no parameters
- Success: 200 with {"status":"ok"}
- Errors: 503 with {"detail":"..."} when readiness checks fail

Auth

GET /v1/auth/nonce

- Auth: none
- Request: query parameter address with a 42-character wallet address
- Success: 200 with {"nonce":"..."}
- Errors: 422 if the address is malformed

Example response:

```
{
  "nonce": "f1d0b88e3f8b4dce"
}
```

POST /v1/auth/verify

- Auth: none
- Request: body with message and signature
- Success: 200 with access_token, refresh_token, and the authenticated account
- Errors: 401 when the signature or message is invalid

Example request:

```
{
  "message": "127.0.0.1 wants you to sign in with your Ethereum
account:\n0x...\n\nURI: http://127.0.0.1:8000\nVersion: 1\nChain ID:
84532\nNonce: f1d0b88e3f8b4dce\nIssued At: 2026-03-17T12:00:00Z",
  "signature": "0x..."
}
```

Example response:

```
{
  "access_token": "eyJ...",
  "refresh_token": "eyJ...",
  "account": {
    "id": 1,
    "wallet_address": "0x...",
    "account_type": "human",
    "is_admin": false,
    "display_name": "Demo Account",
    "created_at": "2026-03-17T12:00:00Z",
    "updated_at": "2026-03-17T12:00:00Z"
  }
}
```

POST /v1/auth/refresh

- Auth: none
- Request: body with `refresh_token`
- Success: `200` with a fresh `access_token`
- Errors: `401` when the refresh token is invalid or expired

POST /v1/auth/api-keys

- Auth: JWT-only
- Request: body with optional `name` and `expires_at`
- Success: `201` with metadata plus the plaintext `api_key` once
- Errors: `422` when the request is invalid

GET /v1/auth/api-keys

- Auth: JWT-only
- Request: no body
- Success: `200` with a list of API-key metadata
- Errors: none expected in the normal path

DELETE /v1/auth/api-keys/{api_key_id}

- Auth: JWT-only
- Request: path parameter `api_key_id`
- Success: `204` with no body
- Errors: `404` when the key does not exist

Account

GET /v1/account/me

- Auth: JWT-only

- Request: no body
- Success: **200** with the current account
- Errors: **404** when the account cannot be found

PATCH /v1/account/me

- Auth: JWT-only
- Request: body with optional **display_name**
- Success: **200** with the updated account
- Errors: **422** when no updatable field is provided or the data is invalid

Example request:

```
{
  "display_name": "Demo Provider"
}
```

POST /v1/account/wallet

- Auth: JWT-only
- Request: body with **wallet_address**
- Success: **200** with a wallet-change nonce and expiry
- Errors: **409** when the requested change conflicts with account state, **422** for invalid input

POST /v1/account/wallet/confirm

- Auth: JWT-only
- Request: body with **message** and **signature**
- Success: **200** with fresh tokens and the updated account
- Errors: **409** when the wallet change flow is not valid or has expired

Provider Authoring

POST /v1/provider/services

- Auth: generic bearer
- Request: body with **slug**, **name**, **summary**, and optional **description**
- Success: **201** with the created service
- Errors: **409** on conflicts, **422** for invalid input

GET /v1/provider/services

- Auth: generic bearer
- Request: no body
- Success: **200** with the authenticated account's services
- Errors: none expected in the normal path

GET /v1/provider/services/{service_id}

- Auth: generic bearer
- Request: path parameter `service_id`
- Success: `200` with the owned service detail
- Errors: `404` when the service does not exist or is not owned

PATCH `/v1/provider/services/{service_id}`

- Auth: generic bearer
- Request: body with optional `name`, `summary`, and `description`
- Success: `200` with the updated service
- Errors: `409` on invalid state transitions, `422` for invalid data

POST `/v1/provider/services/{service_id}/tags`

- Auth: generic bearer
- Request: body with `tags` as the full replacement list
- Success: `200` with the service and updated tags
- Errors: `404` if the service is not owned, `422` for invalid input

POST `/v1/provider/services/{service_id}/publish`

- Auth: generic bearer
- Request: no body
- Success: `200` with the published service
- Errors: `409` when publish requirements are not met

POST `/v1/provider/services/{service_id}/endpoints`

- Auth: generic bearer
- Request: body with endpoint `key`, `name`, `summary`, `description`, `access_mode`, `request_schema`, `response_schema`, `timeout_seconds`, `is_enabled`, and optional `pricing`
- Success: `201` with the created endpoint
- Errors: `409` on invalid service state, `422` for schema or pricing problems

PATCH `/v1/provider/endpoints/{endpoint_id}`

- Auth: generic bearer
- Request: body with optional endpoint fields and optional `pricing`
- Success: `200` with the updated endpoint
- Errors: `404` if the endpoint is not owned, `409` on invalid state transitions, `422` for invalid input

PUT `/v1/provider/endpoints/{endpoint_id}/upstream`

- Auth: generic bearer
- Request: body with `base_url`, `path`, `http_method`, and optional `config`
- Validation: `path` must start with `/` and must not include a scheme, host, query string, or fragment
- Success: `204` with no body
- Errors: `404` if the endpoint is not owned, `409` on invalid state transitions, `422` for invalid input

Example upstream config payload:

```
{
  "base_url": "https://provider.example.com",
  "path": "/invoke",
  "http_method": "POST",
  "config": {
    "auth": {
      "type": "hmac_sha256",
      "key_id": "gateway-key",
      "secret": "gateway-secret"
    }
  }
}
```

Discovery

GET /v1/services

- Auth: none
- Request: no body
- Success: **200** with the public service list
- Errors: none expected in the normal path

GET /v1/services/{service_id_or_slug}

- Auth: none
- Request: path parameter **service_id_or_slug**
- Success: **200** with the public service detail
- Errors: **404** when the service does not exist or is not publicly available
- Lookup rule: a path segment made only of digits is treated as a service id, not a slug

GET /v1/services/{service_id_or_slug}/schema

- Auth: none
- Request: path parameter **service_id_or_slug**
- Success: **200** with the enabled endpoint schemas
- Errors: **404** when the service does not exist or is not publicly available

GET /v1/services/{service_id_or_slug}/pricing

- Auth: none
- Request: path parameter **service_id_or_slug**
- Success: **200** with the enabled endpoint pricing data
- Errors: **404** when the service does not exist or is not publicly available

POST /v1/services/{service_id_or_slug}/quote

- Auth: none
- Request: body with **endpoint_key** and **payload**

- Payload shape: **payload** may be any JSON value that matches the endpoint request schema, not only an object
- Success: **201** with a quote bound to the request payload
- Errors: **404** when the service or endpoint cannot be found, **409** when the quote cannot be created for the current state

Example request:

```
{
  "endpoint_key": "paid-summary",
  "payload": {
    "message": "Summarize this marketplace request."
  }
}
```

Example response:

```
{
  "id": 1,
  "service_id": 1,
  "endpoint_key": "paid-summary",
  "pricing_type": "fixed_per_call",
  "amount_minor": 100,
  "currency": "USD",
  "request_hash":
"0123456789abcdef0123456789abcdef0123456789abcdef",
  "service_revision_id": 1,
  "service_change_token":
"0123456789abcdef0123456789abcdef0123456789abcdef",
  "expires_at": "2026-03-17T12:05:00Z",
  "created_at": "2026-03-17T12:00:00Z"
}
```

Invoke

POST /v1/invoke/{service_id_or_slug}

- Auth: generic bearer
- Request: body with **endpoint_key**, **payload**, and optional **quote_id**
- Payload shape: **payload** may be any JSON value that matches the endpoint request schema, not only an object
- Required header: **Idempotency-Key**
- Success: **200** with the invocation record, or **402** for unpaid paid-endpoint requests
- Errors: **404** for missing services or invocations, **409** for conflicts or in-progress requests, **502** for upstream failures, **504** for upstream timeouts
- Invocation status values: **in_progress**, **succeeded**, **failed**

- Upstream idempotency: the forwarded `X-Agent-Marketplace-Invocation-Id` header is stable across retries for the same invocation and should be treated as the provider-side idempotency key

Example free-invoke request:

```
{
  "endpoint_key": "free-ping",
  "payload": {
    "message": "hello from the free route"
  }
}
```

Example invocation response:

```
{
  "id": 1,
  "service_id": 1,
  "endpoint_key": "free-ping",
  "access_mode": "free",
  "quote_id": null,
  "idempotency_key": "consumer-example-1",
  "request_hash":
  "0123456789abcdef0123456789abcdef0123456789abcdef",
  "status": "succeeded",
  "upstream_status_code": 200,
  "response_payload": {
    "message": "pong",
    "mode": "free"
  },
  "error_message": null,
  "created_at": "2026-03-17T12:00:00Z"
}
```

GET /v1/invocations/{invocation_id}

- Auth: generic bearer
- Request: path parameter `invocation_id`
- Success: `200` with the invocation detail
- Errors: `404` when the invocation is not accessible

GET /v1/invocations

- Auth: generic bearer
- Request: no body
- Success: `200` with the caller's invocation list
- Errors: none expected in the normal path

Finance

GET /v1/provider/earnings

- Auth: generic bearer
- Request: no body
- Success: 200 with earnings totals grouped by currency
- Errors: none expected in the normal path

GET /v1/provider/ledger

- Auth: generic bearer
- Request: no body
- Success: 200 with ledger entries
- Errors: none expected in the normal path

GET /v1/provider/payouts

- Auth: generic bearer
- Request: optional `status` query parameter
- Success: 200 with payout summaries and payout records
- Errors: none expected in the normal path

POST /v1/provider/payouts

- Auth: JWT-only
- Request: no body, but `Idempotency-Key` is required
- Success: 200 with payout-request results
- Errors: 409 when a duplicate or conflicting request is submitted

Example response:

```
{
  "idempotency_key": "example-provider-payout-1",
  "requested_count": 2,
  "sent_count": 2,
  "failed_count": 0,
  "payouts": []
}
```

Admin

POST /v1/admin/services/{service_id}/suspend

- Auth: admin actor
- Request: body with `reason`
- Success: 201 with the moderation action
- Errors: 404 when the service cannot be found, 409 for invalid transitions

POST /v1/admin/services/{service_id}/restore

- Auth: admin actor
- Request: body with **reason**
- Success: **201** with the moderation action
- Errors: **404** when the service cannot be found, **409** for invalid transitions

POST /v1/admin/services/{service_id}/delist

- Auth: admin actor
- Request: body with **reason**
- Success: **201** with the moderation action
- Errors: **404** when the service cannot be found, **409** for invalid transitions

GET /v1/admin/moderation/actions

- Auth: admin actor
- Request: **service_id** query parameter
- Success: **200** with the moderation action history
- Errors: **422** for invalid query values

Design Notes

- Quote creation is public. Consumers do not need to authenticate to request a quote.
- Discovery routes only expose public, active service data.
- Provider upstream URLs and credentials are never returned by discovery responses.
- The paid invoke flow is a two-step **402** challenge/settle flow backed by x402-compatible headers.
- The provider payout request path is separate from payout reporting and requires JWT auth plus idempotency protection.